6.2 Installation
All devices and components should, as fitted in the ship, be designed to

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A complete instruction and maintenance manual with drawings, electric diagrams, spare parts list, etc., should be furnished with each incinerator.
6.2 Installation
All devices and components should, as fitted in the ship, be designed to operate when the ship is upright and when inclined at any angle of list up to and including 15° either way under static conditions and 22.5° under dynamic conditions (rolling) either way and simultaneously inclined dynamically (pitching) 7.5° by bow or stern.
6.3 Incinerator
6.3.1 Incinerators are to be fitted with an energy source with sufficient energy to ensure a safe ignition and complete combustion. The combustion is to take place at sufficient negative pressure in the combustion chamber(s) to ensure no gases or smoke leaking out to the surrounding areas. (See 5.3.1.3)
6.3.2 A driptray is to be fitted under each burner and under any pumps, strainers, etc., that require occasional examination.
7 Tests
7 . 1 Prototype tests
An operating test for the prototype of each design should be conducted, with a test report completed indicating results of all tests. The tests should be

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strainers, etc., that require occasional examination.
7 Tests
7 . 1 Prototype tests
An operating test for the prototype of each design should be conducted, with a test report completed indicating results of all tests. The tests should be conducted to ensure that all of the control components have been properly installed and that all parts of the incinerator, including controls and safety devices, are in satisfactory operating condition. Tests should include those described in section 7.3 below.
7.2 Factory tests
For each unit, if preas sembled, an operating test should be conducted to ensure that all of the control components have been properly installed and that all parts of the incinerator, including controls and safety devices, are in satisfactory operating condition. Tests should include those described in 7.3 below.
7.3 Installation tests
An operating test after installation should be conducted to ensure that all of the control components have been properly installed and . that all parts of the incinerator, including controls and safety devices, are in satisfactory operating condition.
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of the incinerator, including controls and safety devices, are in satisfactory operating condition.
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7.3.1 Flame safeguard. The operation of the flame safeguard system should be verified by causing flame and ignition failures. Operation of the audible alarm (where applicable) and visible indicator should be verified. The shutdown times should be verified.
7.3.2 Limit controls. Shutdown due to the operation of the limit controls should be verified.
7.3.2.1 Oil pressure limit control. The lowering of the fuel oil pressure below the value required for safe combustion should initiate a safety shutdown.
7.3.2.2 Other interlocks. Other interlocks provided should be tested for proper operation as specified by the unit manufacturer.
7.3.3 Combustion controls. The combustion controls should be stable and operate smoothly.
7.3.4 Programming controls. Programming controls should be verified as controlling and cycling the unit in the intended manner. Proper prepurge,

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7.3.3 Combustion controls. The combustion controls should be stable and operate smoothly.
7.3.4 Programming controls. Programming controls should be verified as controlling and cycling the unit in the intended manner. Proper prepurge, ignition, postpurge, and modulation should be verified. A stopwatch should be used for verifying intervals of time.
7.3.5 Fuel supply controls. The satisfactory operation of the two fuel control solenoid valves for all conditions of operation and shutdown should be verified.
7.3.6. Low voltage test. A low voltage test should be conducted to satisfactorily demonstrate that the fuel supply to the burners will be automatically shut off before an incinerator malfunction results from the reduced voltage.
7.3.7 Switches. All switches should be tested to verify proper operation.
8 Certification
8.1 Manufacturer's certification that an incinerator has been constructed in accordance with this standard should be provided (by letter, certificate, or in the instruction manual).
9 Marking
9.1 Each incinerator should be permanently marked indicating:
9.1.1 Manufacturer's name or trademark.

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accordance with this standard should be provided (by letter, certificate, or in the instruction manual).

9 Marking

9.1 Each incinerator should be permanently marked indicating:

9.1.1 Manufacturer's name or trademark.

9.1.2 Style, type, model or other manufacturer's designation for the incinerator.

9.1.3 Capacity - to be indicated by net designed heat release of the incinerator in heat units per timed period; for example, British Thermal Units per hour, megajoules per hour, kilocalories per hour.

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10 Quality assurance

10.1 Incinerators should be designed, manufactured and tested in a manner that ensures they meet the requirements of this standard.

10.2 The incinerator manufacturer should have a quality system that meets ISO 9001, "Quality Systems - Model for Quality Assurance in Design/Development, Production, Installation and Servicing". The quality system should consist of elements necessary to ensure that the production

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meets ISO 9001, "Quality Systems - Model for Quality Assurance in Design/Development, Production, Installation and Servicing". The quality system should consist of elements necessary to ensure that the production quality of the incinerators are designed, tested and marked in accordance with this standard. At no time should an incinerator be sold with this standard designation that does not meet the requirements herein (see Certification).

Al - EMISSION STANDARD FOR SHIPBOARD INCINERATORS WITH CAPACITIES OF UP TO 1,160 kW

Minimum information to be provided

A1.1 An IMO TYPE APPROVAL CERTIFICATE should be required for each shipboard incinerator. In order to obtain such certificate, the incinerator should be designed and built to an IMO approved standard. Each model should go through a specified type approval test operation at the factory or an approved test facility, and under the responsibility of the Administration.

A1.2 TYPE APPROVAL TEST SHOULD INCLUDE MEASURING OF THE FOLLOWING PARAMETERS:

Max capacity

Pilot fuel consumption

02 Average in combustion chamber/zone

CO Average in flue gas

Soot number average

Combustion chamber flue gas

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SOURCE: MEPC.59(33).pdf
A1.2 TYPE APPROVAL TEST SHOULD INCLUDE MEASURING OF THE FOLLOWING PARAMETERS:
Max capacity
Pilot fuel consumption
02 Average
in combustion chamber/zone
CO Average in flue gas
Soot number average
Combustion chamber flue gas
outlet temperature average:
Amount of unburned
components in ashes
Al.3 DURATION OF TEST OPERATION
For sludge oil burning
For solid waste burning
W/9096D/ljt/EWP
kW or kcal/h
kg/h of specified waste
kg/h per burner
kg/h per burner
mg/MJ
Bacharach or ringelman